

Lagrangian Methods and Consumer Demand Examples

Alejandro Melo Ponce



Stony Brook University

September 27, 2016

What can go wrong?

- The condition $MRS = p_x/p_y$ helps identify an interior optimum, under the assumption of differentiability, convexity and monotonicity of preferences.
- There are many important and economically relevant cases for which these assumptions do not hold.
- Non-differentiable preferences: perfect complements.
- No strict convexity: perfect substitutes.
- Non-convex preferences:
 - Preferences with bliss points¹
 - Preferences for extremes²
- More generally, how would we go about solving the problem if instead of just two goods we have 3 or more, i.e. the n good case.
- This class, we will introduce a **very powerful method** that can handle all these cases.
- But first, let's analyze Perfect complements and perfect substitutes.

¹Exercise 6, HW1, Exercise 2, HW2

²Exercise 3, HW3

Perfect Complements

- Consider the following utility function: $u(x, y) = \min\{\alpha x, \beta y\}$, for $\alpha > 0, \beta > 0$.
- Intuition: This utility represents the case, in which you must consume x and y in the fixed proportion $\alpha x = \beta y$.
- The indifference curves as we have seen, look like right angles.
- They have a kink precisely at the set of points $\alpha x = \beta y$. Hence they are not-differentiable.
- Thus we won't be able to use calculus to solve for the optimal points.
- Note that the preferences are monotone, i.e. the more the better. Also they are convex, albeit not strictly convex.
- Since they are monotone, the consumer will exhaust his budget, thus, the budget constraint must hold with equality $p_x x + p_y y = I$.
- Using this and the fact that consumption happens in fixed proportions, $\alpha x = \beta y$ yields two equations in two unknowns which we can solve to get the demands.

Perfect Complements Continued

- We have that $y = \frac{\alpha}{\beta}x$.
- We plug this in the budget constraint to get

$$\begin{aligned}p_x x + p_y \left(\frac{\alpha}{\beta} x \right) &= I \\x \left(p_x + p_y \frac{\alpha}{\beta} \right) &= x \left(\frac{\beta p_x + \alpha p_y}{\beta} \right) = I \\x^M(p_x, p_y, I) &= \frac{\beta I}{\beta p_x + \alpha p_y}; \quad y^M(p_x, p_y, I) = \frac{\alpha I}{\beta p_x + \alpha p_y}\end{aligned}$$

Perfect Substitutes

- Consider the following utility function: $u(x, y) = \alpha x + \beta y$, where $\alpha > 0$ and $\beta > 0$.
- Intuition: This utility represents the case in which goods x and y are perfect substitutes between themselves at the rate $\frac{\alpha}{\beta}$.
- Example: Coke is very similar to Pepsi, but I still prefer coke. Let's say that I would be willing to trade with you 1 bottle of Coke for 2 bottles of Pepsi. Then $u(C, P) = 2C + P$.
- Notice that the MRS is constant: $MRS = \frac{2}{1}$, and represents exactly the trade that I described in the previous bullet.
- These preferences are monotone, so the budget constraint will hold with equality. However, they are not strictly convex.
- We will get **Corner Solutions**.
- Intuitively, how should you allocate your income between coke and pepsi?
- You should choose to consume the good which gives you the more "Bang for your Buck"

Perfect Substitutes Continued

- Clearly is not always the case in which we will have $MRS = \frac{p_x}{p_y}$.
- What should you do?
 - If $MRS > \frac{p_x}{p_y}$ then you should consume all your income on x , because it provides the more satisfaction per dollar spent. Thus $x^M(p_x, p_y, I) = I/p_x, y^M(p_x, p_y, I) = 0$.
 - If $MRS < \frac{p_x}{p_y}$ then you should consume all your income on y , because it provides the more satisfaction per dollar spent. Thus $x^M(p_x, p_y, I) = 0, y^M(p_x, p_y, I) = I/p_y$.
 - If $MRS = \frac{p_x}{p_y}$, then you are indifferent between any x and y that satisfy the budget constraint. Then you should consume $x^M \in [0, I/p_x]$ and $y = \frac{I - p_x x^M}{p_y}$.

Constrained Optimization: The General Approach

- At this point, you should be convinced that a lot of things we do in Economics have to do with Optimization (Maximization or Minimization) and some of these problems are subject to a constraint.
- We have discussed the case of Utility Maximization subject to a budget constraint, but what if the agent is constrained in some other way.
- In general, what should we do if we face a problem with several constraints.
- Another issue: Until now, we have been solving the problem under the assumption that the budget constraint, which is described by an inequality holds with strict equality.
- There are some cases in which the optimum can be found **inside the budget set**, like in Exercise 6 of HW 1.
- Also, assuming all conditions are met, $MRS = p_x/p_y$ is only for the two variable case, so what should we do if we have many variables?

A version of the general problem

- Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $g : \mathbb{R}^n \rightarrow \mathbb{R}$ and $h : \mathbb{R}^n \rightarrow \mathbb{R}$ be multivariate real-valued functions.
- The standard constrained optimization problem is given by:

$$\begin{aligned} \max_{(x_1, \dots, x_n) \in \mathbb{R}^n} : & \quad f(x_1, \dots, x_n) \\ \text{such that} & \quad g(x_1, \dots, x_n) = M \\ & \quad h(x_1, \dots, x_n) \leq Q \end{aligned} \tag{P}$$

- Where M, Q are two real numbers.
- We have two constraints, one that should hold with equality, and one that can either hold with equality or strict inequality at an optimum.
- We can solve this problem with the method of Lagrange.

The Method of Lagrange multipliers

- First consider the version of problem (P) in which we only have the equality constraint.

$$\begin{aligned} \max_{(x_1, \dots, x_n) \in \mathbb{R}^n} : & \quad f(x_1, \dots, x_n) \\ \text{such that} & \quad g(x_1, \dots, x_n) = M \end{aligned} \quad (\text{PE})$$

- We define the **Lagrangian function** as

$$L(x_1, \dots, x_n, \lambda) = f(x_1, \dots, x_n) + \lambda(M - g(x_1, \dots, x_n)) \quad (1)$$

- The Lagrangian function is adding one more variable to the problem λ .
- The meaning of λ will be important for what we study.

Solving the problem

- From here, the rest is very standard, we are going to take first order conditions of the Lagrangian with respect to $x_1, \dots, x_n$ and also λ .

$$[x_1]: f_{x_1}(x_1, \dots, x_n) - \lambda g_{x_1}(x_1, \dots, x_n) = 0$$

$$[x_2]: f_{x_2}(x_1, \dots, x_n) - \lambda g_{x_2}(x_1, \dots, x_n) = 0$$

$$\vdots$$

$$[x_n]: f_{x_n}(x_1, \dots, x_n) - \lambda g_{x_n}(x_1, \dots, x_n) = 0$$

$$[\lambda]: M - g(x_1, \dots, x_n) = 0$$

- We have thus a system of $n + 1$ equations in $n + 1$ unknowns, so it can be solved.

Why the method works?

- λ is measuring the cost of violating the constraint.
- The λ term is acting as a “punishment”, the more you violate the constraint, the lower the value of the lagrangian is going to be.
- Thus, at an optimum, you must be meeting the constraint, because then the term $\lambda(M - g(x_1, \dots, x_n))$ will be equal to zero.
- Another, way to see this, notice that rewriting the conditions, we have

$$\frac{f_{x_1}}{g_{x_1}} = \frac{f_{x_2}}{g_{x_2}} = \dots = \frac{f_{x_n}}{g_{x_n}} = \lambda$$

- You can think of each term as follows: in the numerator we have the **marginal benefit of x_i** , while in the denominator we have the **marginal cost of x_i** .
- Thus, at an optimum the ratio of marginal benefit to marginal cost must be equalized across all variables.

Example: Cobb Douglas Revisited

- Consider the following problem:

$$\max_{(x,y) \in \mathbb{R}^2} : \ln x + \ln y \quad \text{subject to } p_x x + p_y y = I$$

- The Lagrangian is:

$$L(x, y, \lambda) = \ln x + \ln y + \lambda(I - p_x x - p_y y)$$

- FOC's are:

$$\frac{\partial L}{\partial x} = \frac{1}{x^*} - \lambda^* p_x = 0$$

$$\frac{\partial L}{\partial y} = \frac{1}{y^*} - \lambda^* p_y = 0$$

$$\frac{\partial L}{\partial \lambda} = I - p_x x^* - p_y y^* = 0$$

Example continued

- The easiest thing to do here is to first get rid of λ^* : using the first two equations we get

$$\lambda = \frac{1}{x p_x^*} \quad \lambda = \frac{1}{y p_y^*}$$

- Combining them we get: $x^* p_x = y^* p_y$.
- Using the previous result and the restriction $m - p_x x^* - p_y y^* = 0$ yields

$$p_x x^* + p_y \left(x^* \frac{p_x}{p_y} \right) = m \rightarrow x^* \left(p_x + p_y \frac{p_x}{p_y} \right) \\ 2p_x x^* = I \rightarrow x^* = \frac{I}{2p_x} \text{ which also implies } y^* = \frac{I}{2p_y}$$

Meaning of the Lagrange Multiplier

- The utility maximization problem with an equality constraint

$$\begin{aligned} \max_{(x_1, \dots, x_n) \in \mathbb{R}^n} : & \quad U(x_1, \dots, x_n) \\ \text{such that} & \quad p_{x_1}x_1 + p_{x_2}x_2 + \dots + p_{x_n}x_n = I \end{aligned} \quad (\text{PE})$$

- The Lagrangian is

$$L(x_1, \dots, x_n, \lambda) = U(x_1, \dots, x_n) + \lambda(I - p_{x_1}x_1 - p_{x_2}x_2 - \dots - p_{x_n}x_n)$$

- FOC's are:

$$[x_1] : \quad U_{x_1}(x_1, \dots, x_n) - \lambda p_{x_1} = 0$$

$$[x_2] : \quad U_{x_2}(x_1, \dots, x_n) - \lambda p_{x_2} = 0$$

$$\vdots$$

$$[x_n] : \quad U_{x_n}(x_1, \dots, x_n) - \lambda p_{x_n} = 0$$

$$[\lambda] : \quad I - p_{x_1}x_1 - p_{x_2}x_2 - \dots - p_{x_n}x_n = 0$$

The Meaning of the Lagrange Multiplier

- We should have that

$$\frac{U_{x_1}}{p_{x_1}} = \frac{U_{x_2}}{p_{x_2}} = \dots = \frac{U_{x_n}}{p_{x_n}} = \lambda$$

- Thus the Lagrange Multiplier represents the **marginal value of an extra dollar of income**.
- Thus λ represents, how much will you be willing to pay to increase your income by \$1 dollar so that you can distribute optimally across goods.
- λ is called the **shadow** price of the constraint.
- It is interpreted as how much the consumer values an extra dollar of income at an optimal bundle.
- Another way to phrase it, how much the consumer would be willing to pay to be “less punished” by her income restriction by \$1 more dollar.